

90. Backreferences: `\N` aur `\k<name>`

Capturing groups `(...)` ka content sirf result ya replacement me nahi, **pattern ke andar** bhi use kar sakte hain.

Number se: `\N`

Pattern me group ko `\N` se reference karte hain (`N` = group number).

Task: single-quoted `'...'` ya double-quoted `"..."` dono tarah ki quoted strings dhundhna.

Dono quotes ko brackets me daal ke `[ '"](.*)['"]` likhein to mixed quotes (`"..."` ya `'...'`) bhi mil jaate hain. Jaise string `"She's the one!"` me galat match:

```
js
let str = `He said: "She's the one!".`;
str.match(/['"](.*)['"]/g); // "She'
```

Pattern ne opening `"` liya aur text `'` tak khaya jo closing ban gaya.

Fix: closing quote **bilkul wahi** hona chahiye jo opening tha. Opening quote ko capturing group me daalo aur `\1` se reference karo: `(['"])(.*)\1`:

```
js
str.match(/(['"])(.*)\1/g); // "She's the one!"
```

Engine pehla quote `(['"])` dhundh ke uska content yaad rakhta hai (pehla group). Pattern me aage `\1` = "wahi text dhundho jo pehle group me tha". `\2` doosre group ka content, `\3` teesre ka.

Dhyan:

- Group me `(?:)` ho (non-capturing) to use reference nahi kar sakte, engine use yaad nahi rakhta.
- Pattern me `\1`, replacement me `$1`. (Replacement me dollar, pattern me backslash.)

Naam se: `\k<name>`

Kai parentheses ho to naam dena convenient hai. Named group ko `\k<name>` se reference karo:

```
js
```

```
let regexp = /(?!<quote>['"])(.*?)\k<quote>/g;  
`He said: "She's the one!".`.match(regexp); // "She's the one!"
```